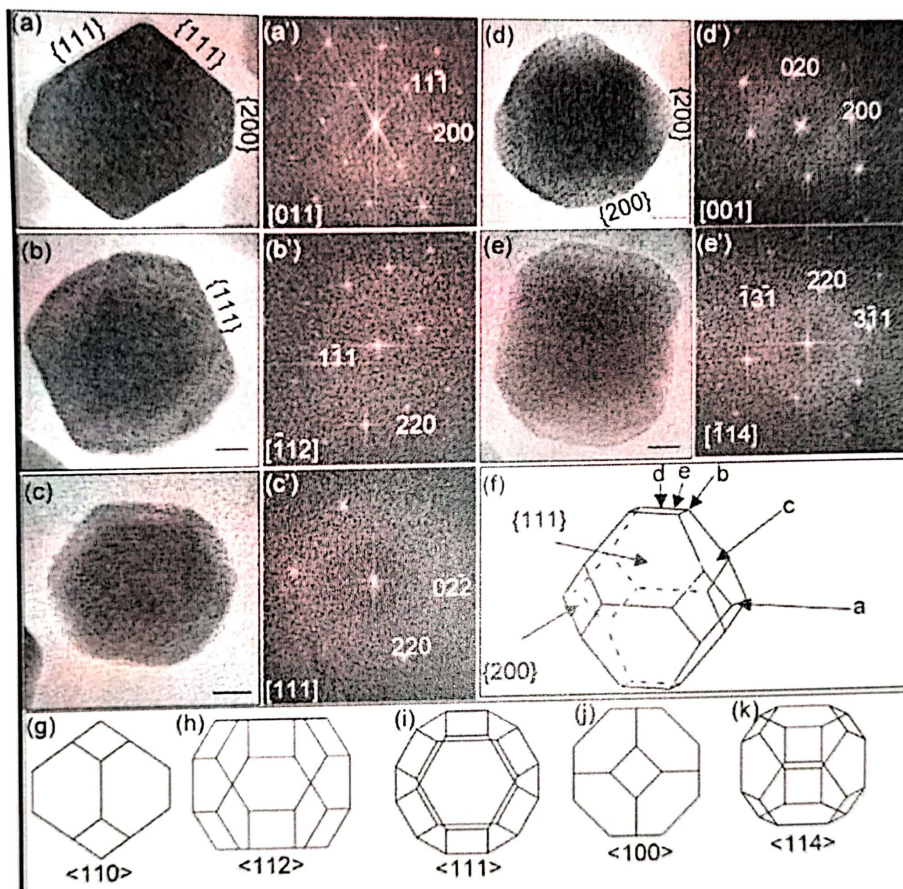


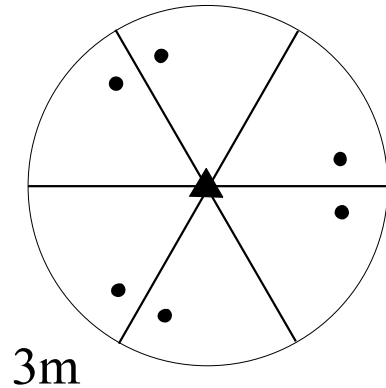
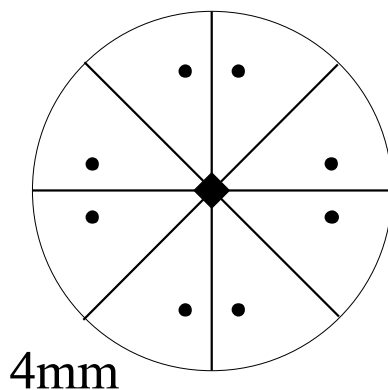
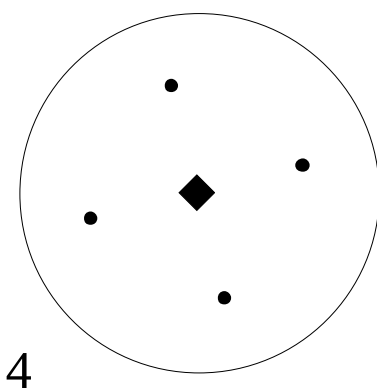
HRTEM images of certain axial directions in silicon crystals

- $[100]$
- $[110]$
- $[111]$
- $[112]$

Exercise: please find the HRTEM images taken along above directions to see 2D patterns formed from the projections of 3D crystal structures.



Exercise: try to draw the stereographic projection of 4, $4mm$, $3m$.

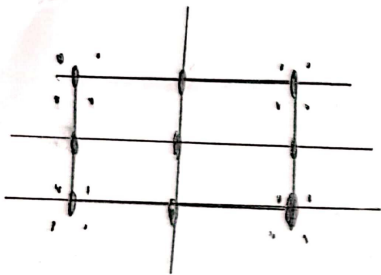


Exercise: Please give the general equivalent points and the distribution of symmetry elements in the following 2D space groups:

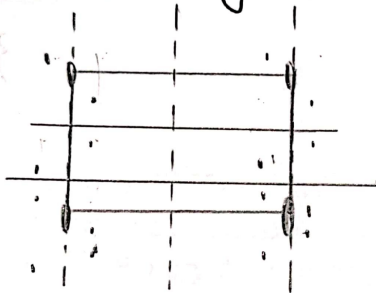
P2mm; P2mg; P2gg ?

Exercise: Please give the general equivalent points and the distribution of symmetry elements in 2D space groups No. 9 2D space group **C2mm**

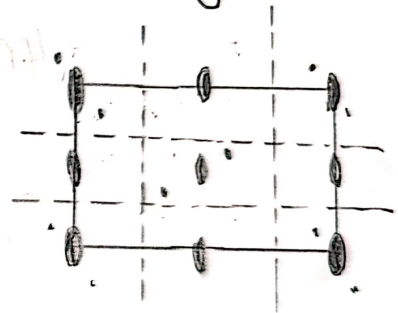
1. P_2mm



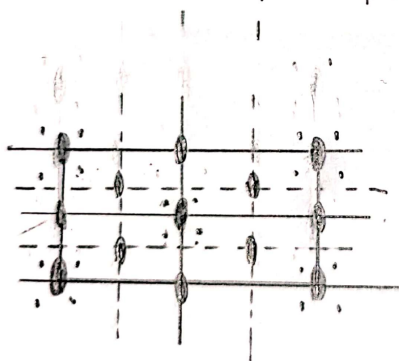
P_2mg



P_2gg



C_2mm



$P1$

$C1?$ no this point group
Only cm , but there is
no m here.

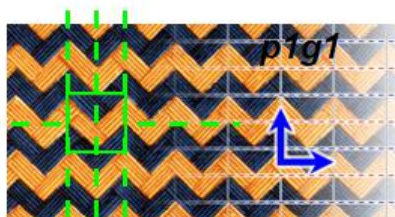


$P211$ or $P4$ (X)

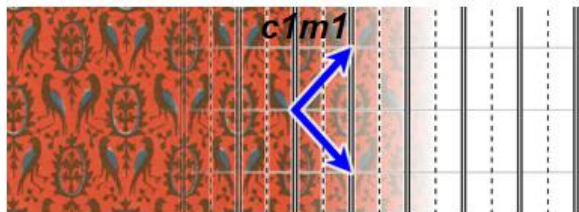


$P1m1$ or $P2mm$ (X)

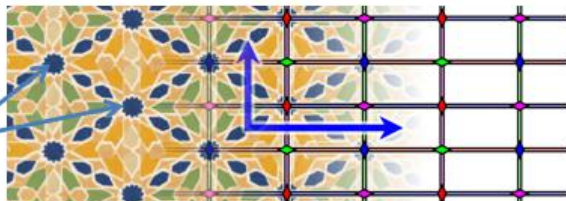
these two points are not
equivalent



$p1g1$

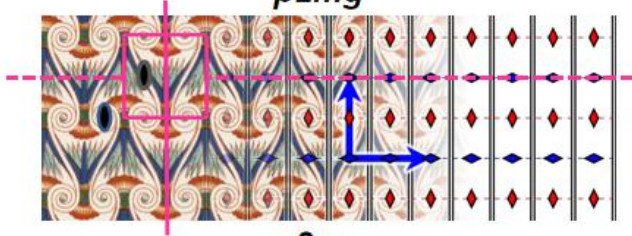


$c1m1$

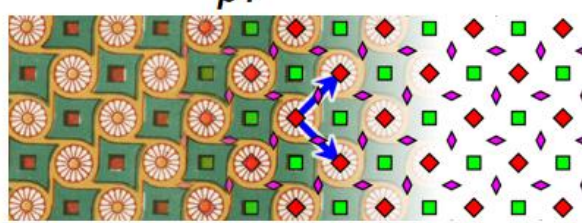


$P2mm$ or $C2mm$ (X)

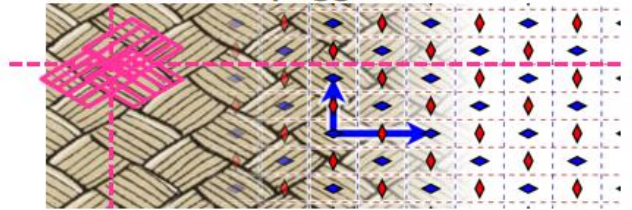
p2mg



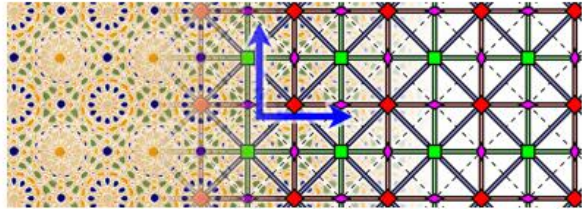
p4



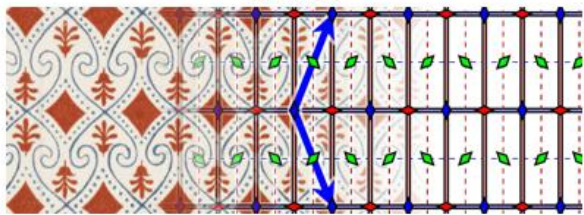
p2gg



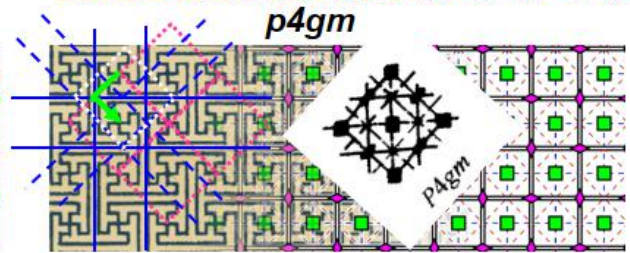
p4mm



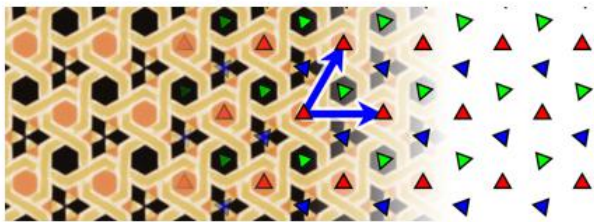
c2mm



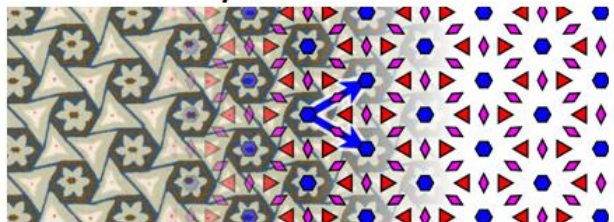
p4gm



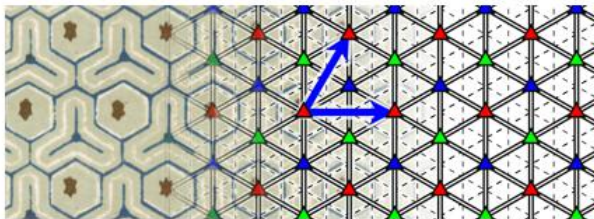
p3



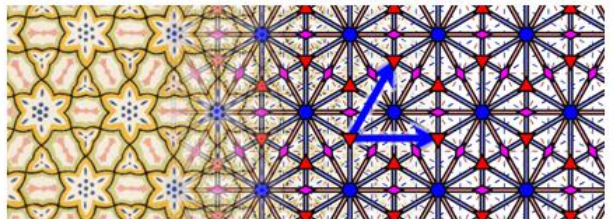
p6



p3m1



p6mm



p31m

